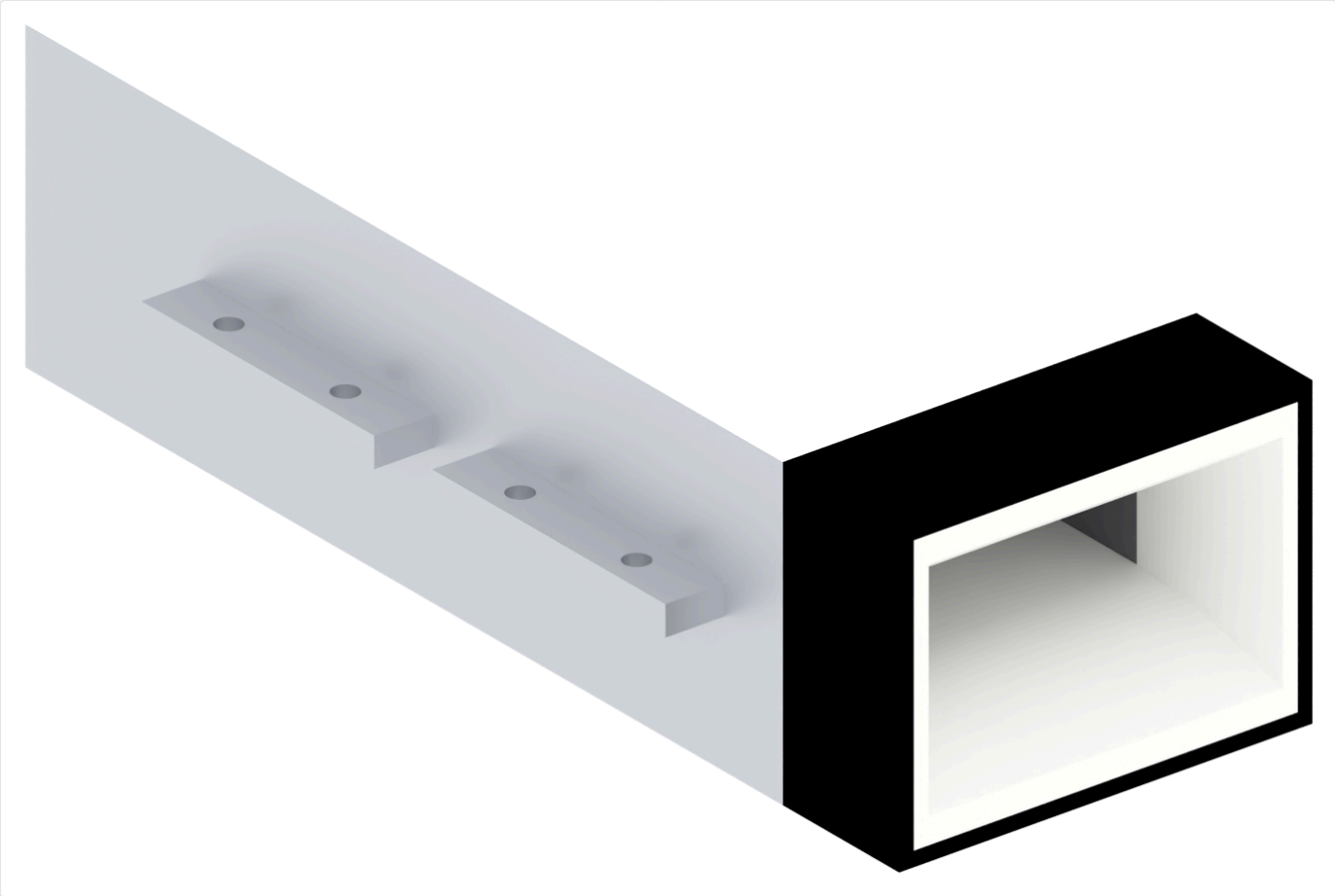
Magnetide v21 — Assembly Guide

Kelas 10.8, Kelompok 1 · MHD seawater pump · 150 × 68 × 41 mm · all dimensions in mm
Generated from the CAD model — every figure below is a render of the delivered STEP file, not a drawing of it.

What you are building

An MHD pump channel: brine (or seawater) runs through a 40 × 14 mm bore, a 483 mT field crosses it, current flows between two graphite plates on the side walls, and the resulting Lorentz force pushes the water out the straight end. A curved bellmouth at the inlet keeps the flow attached so the pump does not cavitate itself.



Finished assembly, inlet (bellmouth) at the near end. Four side lugs per pair carry 8 × M3 bolts that clamp the two steel halves together.

Parts

Printed (PLA or PETG) — from `plastic/`

File	Qty	Size (mm)	Notes
`bellmouth.stl`	1	20 × 56 × 41	flare 48 × 30 → 40 × 14, printed with the mouth facing up

`tray.stl`	2	124 × 52 × 10	magnet tray; one 40 mm pocket on each face
`liner_top.stl` / `liner_bottom.stl`	2	130 × 52 × 1.5	identical — print the same file twice

Metal — from `metal/`

File	Qty	Blank (mm)	Notes
`clamshell-flat.dxf`	2	150 × 110.3	2.0 mm mild steel / SPCC, laser cut, then folded

Bought

Item	Qty	Notes
N52 magnet 20 × 10 × 5	96	24 per face: 6 along × 4 across × 2 deep
Graphite plate 120 × 14 × 2	2	side-wall electrodes
M3 × 16 bolt + nut	8	through the lugs
Epoxy	—	liner to tray, magnets to tray

Key dimensions

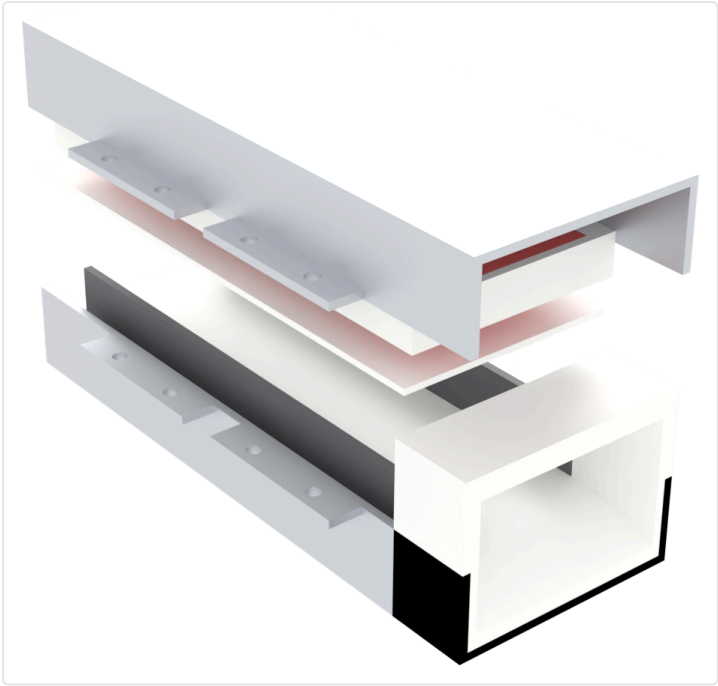
Feature	Value	Why it matters
Channel (bore)	40 × 14	sets current density and Dh = 20.7
Driven length L_e	120	6 stations × 20; force ∝ L
Steel wall	2.0	flux return + structure
Liner	1.5	water barrier over the magnets
Magnetic gap	17	2 × (1.5 liner + 2 × 5 tile) + 2 × 1 air
Bellmouth mouth	48 × 30	flare narrows to the bore over 20 mm
Outlet	straight	a diverging nozzle measurably costs speed — do not add one

Build sequence

1 Fold the two steel halves

Laser-cut two blanks from `clamshell-flat.dxf`. Fold 90° along the dashed bend lines — plate/wall at Y = ±28.0, wall/lug at Y = ±47.15.

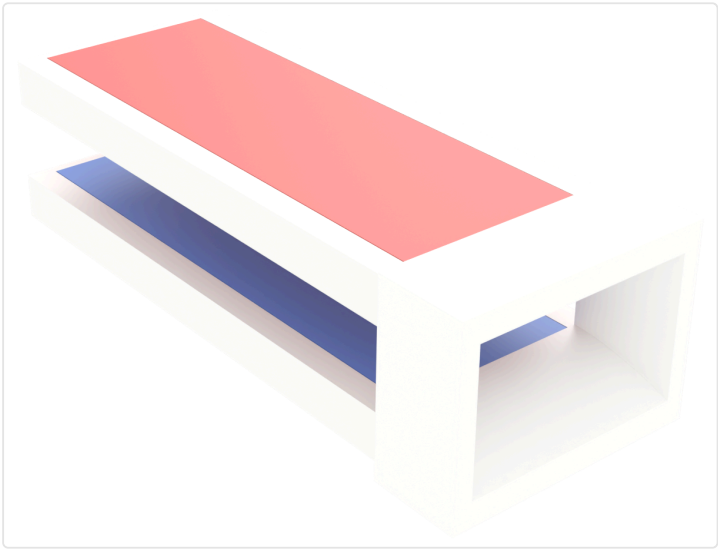
> Bend the first one on a scrap coupon. K = 0.44 / R = 0.2 is nominal for 2 mm mild steel; if your brake differs, the lug holes move and the halves will not line up.



2 Pot the magnets into the trays

Each tray takes 24 tiles: 6 along the length, 4 across the width, 2 deep. Tiles butt edge to edge — the 4 across fill the full 40 mm, which is the whole point: a gap leaves part of the channel un-driven.

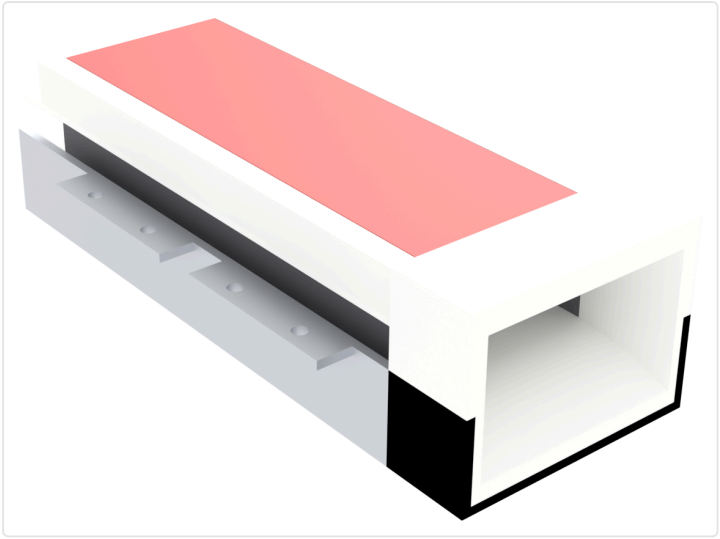
Check polarity station by station with a compass before the epoxy sets. All six stations on one face must point the same way, and the opposite face the other way.



3 Bond the liner onto each tray

1.5 mm plate, epoxied to the magnet face. This is the water barrier — the steel and the magnets must never see brine. Keep the bond line thin; 0.5 mm of epoxy between magnets and water is 0.5 mm of extra magnetic gap.

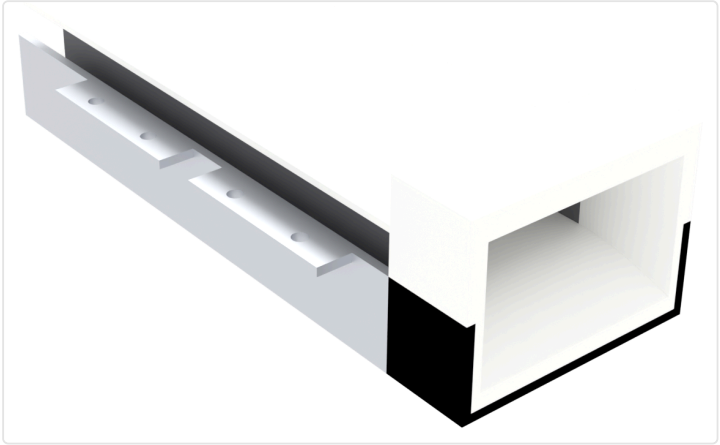
> Nickel-plated N52 corrodes fast in brine. If any tile sits proud of the liner, the pump will fail at that spot, not at the end of the run.



4 Seat the electrodes

Graphite plates 120 × 14 × 2 slide against the 40 mm side walls of the bore, flush with the liner. They are the anode and cathode; keep them clear of the bellmouth so the entry flow is not disturbed.

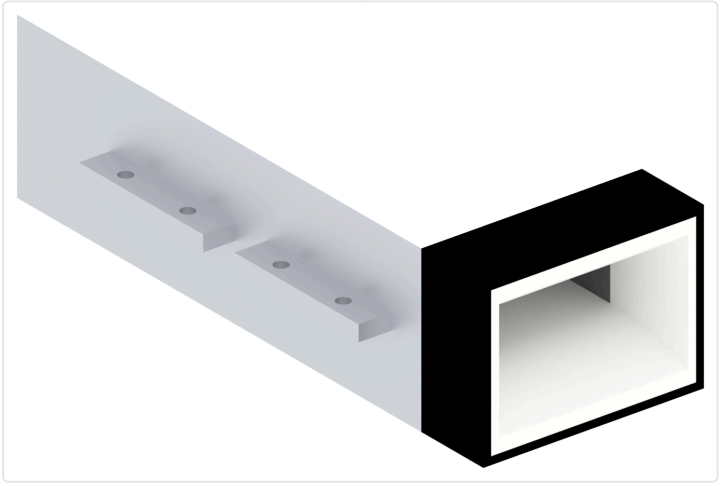
Wire all six stations in parallel. Compass-check the field direction against the current direction once — if they fight, the pump pushes backwards.



5 Close the clamshell

Drop the lower tray assembly into the bottom half, the upper one into the top half, then bring the halves together along the mid-plane. The lug faces are *bare* — no paint, no gasket between them — because they are the flux return path.

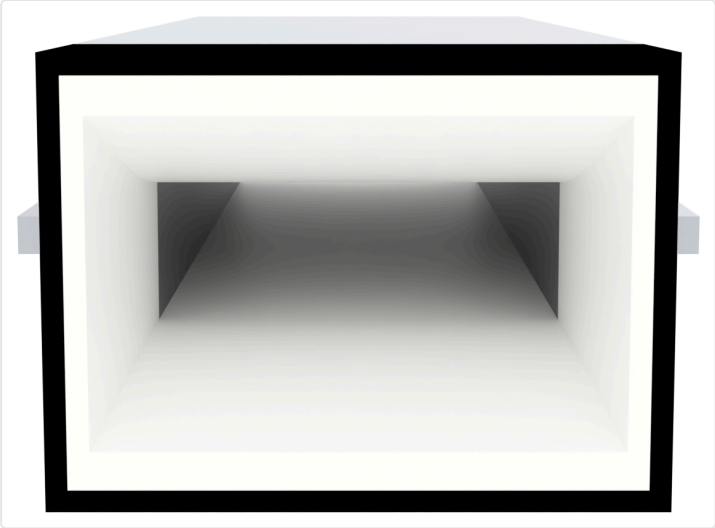
Fit 8 × M3 bolts and tighten in a cross pattern, a little at a time, so the halves seat flat instead of rocking.



6 Bolt on the bellmouth

The printed inlet bolts to the front of the steel tube. Its curved flare takes the 48 × 30 mouth down to the 40 × 14 bore over 20 mm — this is what keeps the entry flow attached.

Leave the outlet **straight**. A diverging nozzle was measured to drop boat speed from 0.3 to 0.2 m/s through recirculation.



Fit and clearance

Joint	Clearance	Consequence if wrong
magnet in tray pocket	0.2 per side	too tight cracks tiles on insertion
tray inside steel cavity	0 (clamped)	slack tray rattles and wears the liner
liner to tray face	0.5 bond line	thick epoxy = wider gap = weaker field
electrode to liner pocket	0.1 per side	leaks past the electrode, current bypasses the water
M3 bolt in lug hole	Ø3.4 hole	standard M3 clearance

Before you power it

- ☐ All 8 bolts seated; halves flush at the mid-plane, no visible gap
- ☐ Compass check: every station on a face same direction, faces opposed
- ☐ Liner unbroken over every magnet pocket — no nickel showing
- ☐ Electrodes flush, not shorting to the steel through a scratch
- ☐ Bore clear: shine a light down the inlet, no epoxy drips in the channel
- ☐ Fuse in the supply line; keep runs under 2 minutes and ventilate (Cl_2)

> **Model status.** Dimensions above come from the shipped STEP file and were measured back off the solid — not copied from the spec. The magnets are modelled as eight 120 mm bars rather than 96 loose tiles: same envelope and same 40 mm coverage, easier to look at, but you cannot count individual tiles in these renders.
